CML — TKI Armory: Drug Selection & Toxicities

1. Imatinib (1st gen)

WHAT YOU SEE

IMATINIB banner over a sword, 1x potency, 400 mg, all phases, $ generic.

WHAT IT ENCODES

Imatinib 400 mg/day is the prototype BCR::ABL1 TKI, baseline potency '1x'. Cheap (generic), well tolerated, approved across phases. Classic side effects: periorbital edema, fluid retention, muscle cramps, rash, cytopenias.

BOARD TRAP

Imatinib is first-line for standard-risk and the default when cost matters; high-risk score or a goal of TFR favors a more potent 2nd-gen agent up front.

2. Dasatinib (~300x)

WHAT YOU SEE

DASATINIB, 300x, 100 mg / 50 mg, lungs, AVOID LUNG DX, 10–25%.

WHAT IT ENCODES

Dasatinib (100 mg daily) is ~300x more potent than imatinib; also a SRC inhibitor. Signature toxicity is pleural effusion (10–25%) and pulmonary arterial hypertension. Penetrates CNS — useful in lymphoid blast crisis.

BOARD TRAP

Avoid dasatinib in patients with pleural/pulmonary disease — pleural effusion is its hallmark adverse effect, not a class effect of all TKIs.

3. Nilotinib (~30x)

WHAT YOU SEE

NILOTINIB, 30x, 300 BID, AVOID DM/CVD, 25–35% 10-yr DM/CVD.

WHAT IT ENCODES

Nilotinib (300 mg BID first-line) is ~30x imatinib potency. Causes hyperglycemia, dyslipidemia and arterial/cardiovascular occlusive events, plus QT prolongation. Take fasting; food increases levels and QT risk.

BOARD TRAP

Avoid nilotinib in diabetics and those with vascular/CV disease — its peripheral arterial occlusive disease and QT prolongation are the discriminators versus dasatinib's pleural effusions.

4. Bosutinib (30–50x)

WHAT YOU SEE

BOSUTINIB, 30–50x, 400 mg, GI/liver icons, AVOID GI/LIVER.

WHAT IT ENCODES

Bosutinib (~400 mg daily) is a SRC/ABL inhibitor with prominent GI toxicity (diarrhea) and transaminase elevations. A good option when cardiopulmonary comorbidities preclude dasatinib/nilotinib.

BOARD TRAP

Bosutinib's limiting toxicity is GI/hepatic — choose it to AVOID the cardiac/pulmonary/vascular profiles of the other 2nd-gen drugs.

5. Ponatinib (45→15 mg)

WHAT YOU SEE

PONATINIB, 45→15 mg, HEAVY HAZARDS, HIGHEST CV, pancreas, 50–60% / 10–20%.

WHAT IT ENCODES

Ponatinib is the 3rd-gen TKI active against T315I. It carries the highest risk of arterial thrombotic/vascular occlusive events and pancreatitis; dose is reduced (e.g., 45→15 mg) once response is achieved to limit toxicity.

BOARD TRAP

Ponatinib is the go-to for T315I-mutated disease but its severe vascular/CV toxicity means it is reserved, not first-line — never start it casually for resistance without checking the mutation.

6. T315I gatekeeper mutation

WHAT YOU SEE

Knight with shield labeled T315I blocking arrows at the ATP target.

WHAT IT ENCODES

T315I is the 'gatekeeper' ATP-pocket mutation conferring resistance to all 1st/2nd-gen TKIs. Only ponatinib and asciminib (and omacetaxine) retain activity.

BOARD TRAP

If a patient fails 2nd-gen TKI, the single most important test is BCR::ABL1 kinase-domain mutation analysis — T315I redirects therapy to ponatinib or asciminib.

7. Asciminib (STAMP/myristoyl)

WHAT YOU SEE

ASCIMINIB boomerang, MYRISTOYL, STAMP, 40 BID / 200 BID T315I.

WHAT IT ENCODES

Asciminib is a first-in-class STAMP inhibitor that binds the myristoyl pocket allosterically rather than the ATP site (40 mg BID; 200 mg BID for T315I). Better tolerated, active against many resistance mutations.

BOARD TRAP

Because asciminib is allosteric (myristoyl pocket), it sidesteps ATP-site mutations — that mechanism is why it works in T315I where 2nd-gen ATP-competitive TKIs fail.

8. Omacetaxine (not a TKI)

WHAT YOU SEE

OMACETAXINE, NOT A TKI, ≥2 TKI FAILURE.

WHAT IT ENCODES

Omacetaxine (homoharringtonine) is a protein-synthesis inhibitor, NOT a TKI; it is independent of BCR::ABL1 kinase mutation status and used after ≥2 TKI failures, including T315I.

BOARD TRAP

Omacetaxine has a mechanism unrelated to the kinase, so kinase-domain mutations (even T315I) do not confer resistance to it.

9. Treatment goal algorithm

WHAT YOU SEE

Scroll: SURVIVAL → imatinib generic; TFR → 2nd gen; HIGH RISK → 2nd gen.

WHAT IT ENCODES

Goal-directed selection: if the goal is survival/cost, imatinib generic suffices; if the goal is treatment-free remission or the patient is high-risk (high ELTS/Sokal), start a more potent 2nd-generation TKI to reach deep molecular response faster.

BOARD TRAP

More potent does not mean longer life for standard-risk patients — 2nd-gen agents give faster/deeper responses (better TFR odds) but imatinib matches them on overall survival in low-risk disease.